

POWERFACTORY

PowerFactory 2024

Technical Reference

Q(V)-Curve

IntQvcurve

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 General Description

The Q(V)-Curve object (*IntQvcurve*) is used to define for static generators, station controllers and for loads the behaviour of the reactive power depending on the voltage.

The curve can be entered in generator orientation or load orientation.

2 Configuration

The Q(V)-Curve object can be configured at the Configuration page as follow:

- Droop in Mvar/p.u. and Q limits in Mvar
- Droop in % and Q limits in p.u.

When the droop is defined in % and the limits in p.u. the corresponding rated power is used to convert this to Mvar/p.u. and the corresponding limits in Mvar, where then the droop in Mvar/p.u. is calculated as:

$$Qdroop = \frac{S_r \cdot ngnum \cdot 100}{ddroop} \tag{1}$$

where S_r is the rated power of the machine, load or station controller. The $ngnum$ is the no. of parallel units e.g. for static generators.

Please note, that the curve is defined for one machine. So when the no. of parallel units is greater one, the droop must be multiplied by the no. of parallel units ($ngnum$) also in case of the Q(V)-Curve is defined in Mvar/p.u.. Also the Q_{min} and Q_{max} are defined for one machine.

3 Droop Orientation

The droop can be defined as

- Generator Oriented
- Load Oriented

Typically the reactive power infeed is decreased when the voltage increased for generator orient definition and in opposite direction for load orientation. Note for both cases the entered droop should be entered always positive.

3.1 Generator Oriented

If the droop is configured as generator oriented, as higher the voltage es lower is the reactive power infeed. The reactive power limits (Q_{min} and Q_{max}) are also defined in generator orientation.

When the Q(V)-Curve is selected for a generator, the behaviour is exactly like shown in the diagram. In case a generator oriented curve is used for generators the droop is rotated, like

when switched to load oriented, but in addition the Q_{min} and Q_{max} are also rotated ($Q_{min} = -Q_{max}$ and $Q_{max} = -Q_{min}$).

See for example figure 4.1 and figure 4.2.

3.2 Load Oriented

If the droop is configured as load oriented, as higher the voltage es higher is the consumed reactive power. The reactive power limits (Q_{min} and Q_{max}) are also defined in load orientation.

When the Q(V)-Curve is selected for a load, the behaviour is exactly like shown in the diagram. In case a load oriented curve is used for generators the droop is rotated, like when switched to generator oriented, but in addition the Q_{min} and Q_{max} are also rotated ($Q_{min} = -Q_{max}$ and $Q_{max} = -Q_{min}$).

See for example figure 5.1 and figure 5.2.

4 Characteristic Type

The Q(V)-Curve can be configured as curve with or without voltage deadband.

4.0.1 Curve with voltage deadband

When the curve is configured with voltage deadband, a lower and upper voltage limit must be defined where $Q_{ref} = 0$ Mvar.

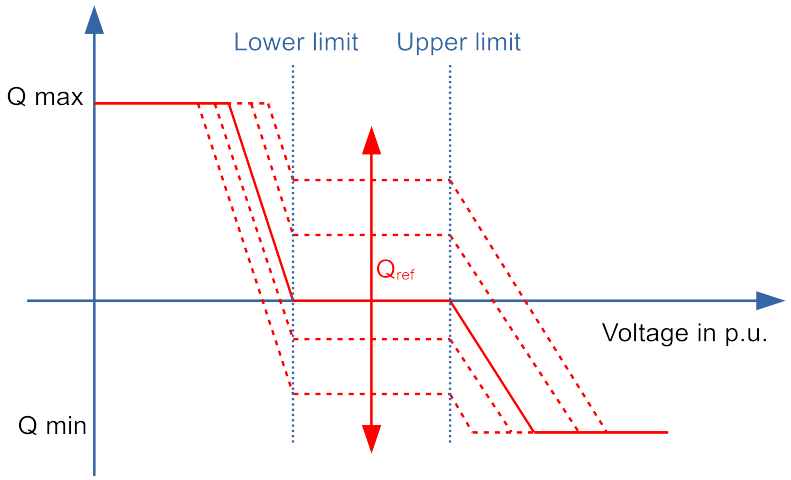


Figure 4.1: With voltage deadband (Generator oriented)

The curve can be shifted horizontally in the corresponding elements by entering the Q_{ref} parameter:

- Static Generator: Q_{ref} can be defined via dispatched reactive power parameter (*qgini* in Mvar)

- Load Model: Q_{ref} can be defined via Q reference parmeter (*Q0qvref* in Mvar or *q0qvref* in p.u.)
- Station Controller: Q_{ref} can be defined by the Q setpoint parameter (*qset* in Mvar)

Note: Q_{ref} must be low than Q_{max} and greater than Q_{min}.

4.0.2 Curve without voltage deadband

The curve is defined, that for 1 p.u. voltage (reference voltage) no reactive power is infeed or consumed ($Q = 0$ Mvar).

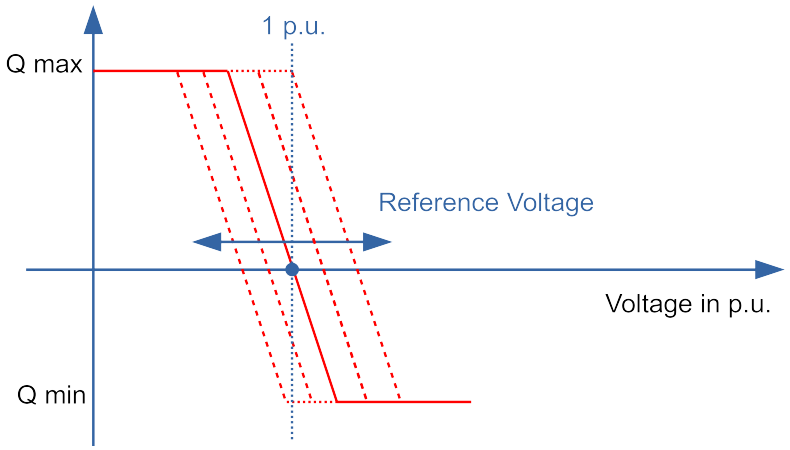


Figure 4.2: Without voltage deadband (Generator oriented)

The curve can be shifted vertically in the corresponding elements by entering the reference voltage (u_{ref}) parameter:

- Static Generator: u_{ref} can be defined via reference voltage parameter (*u0qvref* in p.u.).
- Load Model: u_{ref} can be defined via reference voltage parameter (*u0qvref* in p.u.).
- Station Controller: u_{ref} can be defined by the reference voltage parameter (*u0qvref* in p.u.).

Note: The reference voltage parameter (*u0qvref*) is internally mapped to the voltage setpoint parameter (*usetp*) for static generators and for station controllers.

5 Separate Droop Values for over- and underexcited operation

Different droop values can be entered for the over- and underexcited operation when the option is enabled. An example is shown in the figure below:

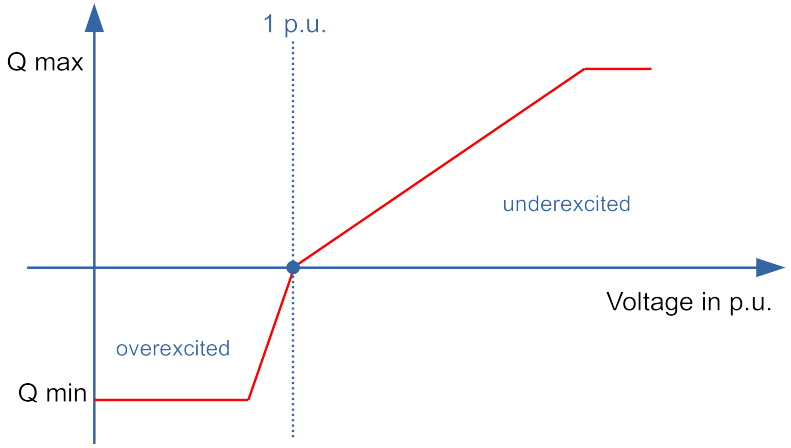


Figure 5.1: Without voltage deadband and different droops (Load oriented)

and an example with deadband:

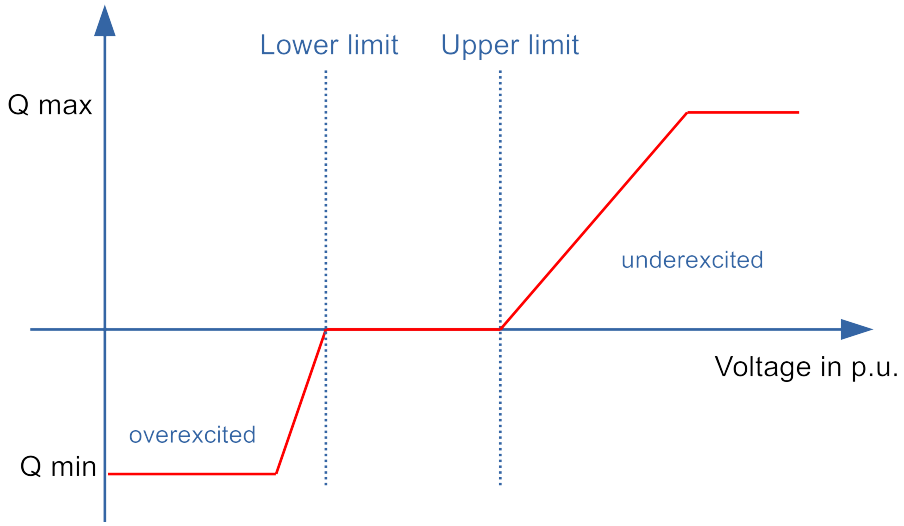


Figure 5.2: With voltage deadband and different droops (Load oriented)

6 Application

The Q(V)-Curve object can be selected in different elements when control mode is set to “Q(V)-Characteristic”. The curve is only used for the load flow model (balanced, unbalanced AC load flow).

The following elements are supporting Q(V)-Curve:

- Static generator, 3 phase and single phase model
- PV System, 3 phase and single phase model
- Station controller
- General load, LV load model (3 phase and single phase model)